Scale Exercises - Section III, Level 23

Three notes forward, one degree back.

Natural Minor Three-Note Rolling Sequence: One-Octave Neck Ladder

Learn the overlap pattern in one octave before extending it across the full six-string frame.

1. Overlap the minor scale

Play 1-2-b3, then begin the next group on degree 2. The line moves forward while revisiting the previous two notes.

2. Preserve the pick cycle

Each triplet has three notes, so its starting stroke alternates. Continue through group and string boundaries without a reset.

3. Shift only after the cell

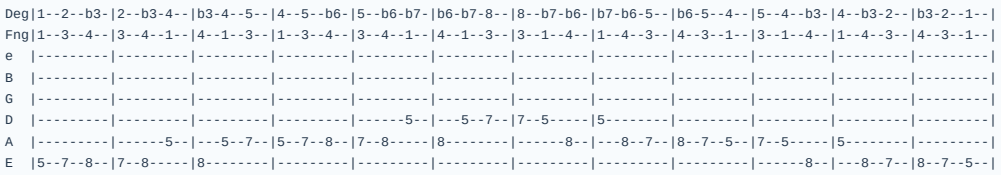
Complete all 12 triplets, return to the low root, release pressure, and move the complete frame exactly one fret.

Worked example - A natural minor at low-E fret 5

Scale notes: A B C D E F G A. The octave root is repeated at the turn.

UP **1-2-b3 | 2-b3-4 | b3-4-5 | 4-5-b6 | 5-b6-b7 | b6-b7-8**

DOWN **8-b7-b6 | b7-b6-5 | b6-5-4 | 5-4-b3 | 4-b3-2 | b3-2-1**



CROSSINGS UP: EEE | EEA | EAA | AAA | AAD | ADD DOWN: DDA | DAA | AAA | AAE | AEE | EEE

KEY ROUTE A -> Bb -> B -> C -> C# -> D -> Eb -> E -> F -> F# -> G -> G# -> A, then reverse.

WHY ONE OCTAVE Level 23 isolates the new rolling motion; Level 24 will expand it across six strings.

Session protocol

1. Say the natural-minor formula and identify b3, b6, and b7.
2. Master the A-minor 36-note cell before adding a shift.
3. Start 12-18 BPM below your clean Level 22 tempo.
4. Use 2-3 neighboring rungs and pause one beat after each cell.
5. Repeat with a downstroke start and an upstroke start.
6. Remove the pause, connect the route, and reverse at fret 17.

Level 23 - Chromatic Natural-Minor Route and Sequence Map

Ascending route: complete one 36-note rolling cell, then move the frame +1 fret

Descending route: use the same cards in reverse order and move -1 fret

Rung	Key	Low-E root	Frame	Scale tones (degrees 1-b7)
01	A natural minor	fret 5	5-8	A B C D E F G
02	Bb natural minor	fret 6	6-9	Bb C Db Eb F Gb Ab
03	B natural minor	fret 7	7-10	B C# D E F# G A
04	C natural minor	fret 8	8-11	C D Eb F G Ab Bb
05	C# natural minor	fret 9	9-12	C# D# E F# G# A B
06	D natural minor	fret 10	10-13	D E F G A Bb C
07	Eb natural minor	fret 11	11-14	Eb F Gb Ab Bb Cb Db
08	E natural minor	fret 12	12-15	E F# G A B C D
09	F natural minor	fret 13	13-16	F G Ab Bb C Db Eb
10	F# natural minor	fret 14	14-17	F# G# A B C# D E
11	G natural minor	fret 15	15-18	G A Bb C D Eb F
12	G# natural minor	fret 16	16-19	G# A# B C# D# E F#
13	A natural minor - octave	fret 17	17-20	A B C D E F G

How to use the map

Up pass:
rungs 01-13.

Turn:
finish the high-A cell at fret 17, then reverse.

Down pass:
rungs 12-01. Skip the high-A card unless starting a new round trip.

Key names:
C#, F#, and G# minor keep the practical sharp spellings.

Goal:
hear the same natural-minor formula while the rolling line rises and falls chromatically.

Sequence anatomy

UP: 1-2-b3 | 2-b3-4 | b3-4-5 | 4-5-b6 | 5-b6-b7 | b6-b7-8
DOWN: 8-b7-b6 | b7-b6-5 | b6-5-4 | 5-4-b3 | 4-b3-2 | b3-2-1

One group per beat as eighth-note triplets. The starting pick stroke alternates at every group, but the 36-note cell preserves the starting stroke for the next key.

Mastery gate before Level 24

Complete the full fret-5 to fret-17 to fret-5 route with a true triplet pulse, silent one-fret shifts, correct early and late crossings, and both pick starts on two separate days.

ROUND-TRIP COUNT: 25 cells x 36 notes = 900 picked notes.

Level 23 - Natural Minor Three-Note Rolling Sequence

Rungs 01-03: play each card as one 12-triplet cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

01 A NATURAL MINOR

ROOT E5 | FRAME 5-8

36-NOTE / 12-TRIPLET CELL

Scale tones: A B C D E F G A

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------5--|---5--7--|7--5-----|5--------|---------|---------|---------|---------|
A  |---------|------5--|---5--7--|5--7--8--|7--8-----|8--------|------8--|---8--7--|8--7--5--|7--5-----|5--------|---------|
E  |5--7--8--|7--8-----|8--------|---------|---------|---------|---------|---------|---------|------8--|---8--7--|8--7--5--|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

02 BB NATURAL MINOR

ROOT E6 | FRAME 6-9

36-NOTE / 12-TRIPLET CELL

Scale tones: Bb C Db Eb F Gb Ab Bb

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------6--|---6--8--|8--6-----|6--------|---------|---------|---------|---------|
A  |---------|------6--|---6--8--|6--8--9--|8--9-----|9--------|------9--|---9--8--|9--8--6--|8--6-----|6--------|---------|
E  |6--8--9--|8--9-----|9--------|---------|---------|---------|---------|---------|---------|------9--|---9--8--|9--8--6--|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

03 B NATURAL MINOR

ROOT E7 | FRAME 7-10

36-NOTE / 12-TRIPLET CELL

Scale tones: B C# D E F# G A B

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------7--|---7--9--|9--7-----|7--------|---------|---------|---------|---------|
A  |---------|------7--|---7--9--|7--9--10-|9--10----|10-------|------10-|---10-9--|10-9--7--|9--7-----|7--------|---------|
E  |7--9--10-|9--10----|10-------|---------|---------|---------|---------|---------|---------|------10-|---10-9--|10-9--7--|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

Level 23 - Natural Minor Three-Note Rolling Sequence

Rungs 04-06: play each card as one 12-triplet cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

04 C NATURAL MINOR

ROOT E8 | FRAME 8-11

36-NOTE / 12-TRIPLET CELL

Scale tones: C D Eb F G Ab Bb C

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------8--|---8--10-|10-8-----|8--------|---------|---------|---------|---------|
A  |---------|------8--|---8--10-|8--10-11-|10-11----|11-------|------11-|---11-10-|11-10-8--|10-8-----|8--------|---------|
E  |8--10-11-|10-11----|11-------|---------|---------|---------|---------|---------|---------|------11-|---11-10-|11-10-8--|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

05 C# NATURAL MINOR

ROOT E9 | FRAME 9-12

36-NOTE / 12-TRIPLET CELL

Scale tones: C# D# E F# G# A B C#

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------9--|---9--11-|11-9-----|9--------|---------|---------|---------|---------|
A  |---------|------9--|---9--11-|9--11-12-|11-12----|12-------|------12-|---12-11-|12-11-9--|11-9-----|9--------|---------|
E  |9--11-12-|11-12----|12-------|---------|---------|---------|---------|---------|---------|------12-|---12-11-|12-11-9--|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

06 D NATURAL MINOR

ROOT E10 | FRAME 10-13

36-NOTE / 12-TRIPLET CELL

Scale tones: D E F G A Bb C D

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------10-|---10-12-|12-10----|10-------|---------|---------|---------|---------|
A  |---------|------10-|---10-12-|10-12-13-|12-13----|13-------|------13-|---13-12-|13-12-10-|12-10----|10-------|---------|
E  |10-12-13-|12-13----|13-------|---------|---------|---------|---------|---------|---------|------13-|---13-12-|13-12-10-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

Level 23 - Natural Minor Three-Note Rolling Sequence

Rungs 07-09: play each card as one 12-triplet cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

07 EB NATURAL MINOR

ROOT E11 | FRAME 11-14

36-NOTE / 12-TRIPLET CELL

Scale tones: Eb F Gb Ab Bb Cb Db Eb

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------11-|---11-13-|13-11----|11-------|---------|---------|---------|---------|
A  |---------|------11-|---11-13-|11-13-14-|13-14----|14-------|------14-|---14-13-|14-13-11-|13-11----|11-------|---------|
E  |11-13-14-|13-14----|14-------|---------|---------|---------|---------|---------|---------|------14-|---14-13-|14-13-11-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

08 E NATURAL MINOR

ROOT E12 | FRAME 12-15

36-NOTE / 12-TRIPLET CELL

Scale tones: E F# G A B C D E

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------12-|---12-14-|14-12----|12-------|---------|---------|---------|---------|
A  |---------|------12-|---12-14-|12-14-15-|14-15----|15-------|------15-|---15-14-|15-14-12-|14-12----|12-------|---------|
E  |12-14-15-|14-15----|15-------|---------|---------|---------|---------|---------|---------|------15-|---15-14-|15-14-12-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

09 F NATURAL MINOR

ROOT E13 | FRAME 13-16

36-NOTE / 12-TRIPLET CELL

Scale tones: F G Ab Bb C Db Eb F

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------13-|---13-15-|15-13----|13-------|---------|---------|---------|---------|
A  |---------|------13-|---13-15-|13-15-16-|15-16----|16-------|------16-|---16-15-|16-15-13-|15-13----|13-------|---------|
E  |13-15-16-|15-16----|16-------|---------|---------|---------|---------|---------|---------|------16-|---16-15-|16-15-13-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

Level 23 - Natural Minor Three-Note Rolling Sequence

Rungs 10-12: play each card as one 12-triplet cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

10 F# NATURAL MINOR

ROOT E14 | FRAME 14-17

36-NOTE / 12-TRIPLET CELL

Scale tones: F# G# A B C# D E F#

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------14-|---14-16-|16-14----|14-------|---------|---------|---------|---------|
A  |---------|------14-|---14-16-|14-16-17-|16-17----|17-------|------17-|---17-16-|17-16-14-|16-14----|14-------|---------|
E  |14-16-17-|16-17----|17-------|---------|---------|---------|---------|---------|---------|------17-|---17-16-|17-16-14-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

11 G NATURAL MINOR

ROOT E15 | FRAME 15-18

36-NOTE / 12-TRIPLET CELL

Scale tones: G A Bb C D Eb F G

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------15-|---15-17-|17-15----|15-------|---------|---------|---------|---------|
A  |---------|------15-|---15-17-|15-17-18-|17-18----|18-------|------18-|---18-17-|18-17-15-|17-15----|15-------|---------|
E  |15-17-18-|17-18----|18-------|---------|---------|---------|---------|---------|---------|------18-|---18-17-|18-17-15-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

12 G# NATURAL MINOR

ROOT E16 | FRAME 16-19

36-NOTE / 12-TRIPLET CELL

Scale tones: G# A# B C# D# E F# G#

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------16-|---16-18-|18-16----|16-------|---------|---------|---------|---------|
A  |---------|------16-|---16-18-|16-18-19-|18-19----|19-------|------19-|---19-18-|19-18-16-|18-16----|16-------|---------|
E  |16-18-19-|18-19----|19-------|---------|---------|---------|---------|---------|---------|------19-|---19-18-|19-18-16-|
```

Finish on the low root; release pressure; move the entire frame one fret.

Continuous alternate picking - D-start and U-start

Level 23 - Final Rung and Mastery Work

Rung 13: complete the octave A-minor cell, then reverse the neck route

13 A NATURAL MINOR - OCTAVE

ROOT E17 | FRAME 17-20

36-NOTE / 12-TRIPLET CELL

Scale tones: A B C D E F G A

```
Deg|1--2--b3-|2--b3-4--|b3-4--5--|4--5--b6-|5--b6-b7-|b6-b7-8--|8--b7-b6-|b7-b6-5--|b6-5--4--|5--4--b3-|4--b3-2--|b3-2--1--|
Fng|1--3--4--|3--4--1--|4--1--3--|1--3--4--|3--4--1--|4--1--3--|3--1--4--|1--4--3--|4--3--1--|3--1--4--|1--4--3--|4--3--1--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
G  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
D  |---------|---------|---------|---------|------17-|---17-19-|19-17----|17-------|---------|---------|---------|---------|
A  |---------|------17-|---17-19-|17-19-20-|19-20----|20-------|------20-|---20-19-|20-19-17-|19-17----|17-------|---------|
E  |17-19-20-|19-20----|20-------|---------|---------|---------|---------|---------|---------|------20-|---20-19-|20-19-17-|
```

Finish on the low root; reverse direction and continue with rung 12.

Continuous alternate picking - D-start and U-start

Full round-trip challenge

1. Start at A natural minor, low-E fret 5.
2. Play rungs 01-13 without restarting the pick cycle.
3. Reverse after the fret-17 cell and return through 12-01.
4. Name each key and say the minor formula before its root.
5. Repeat the route with the opposite starting stroke.

25 cells x 36 notes = 900 picked notes.

After the basic ladder is clean

1. Accent the first note of each triplet, then remove it.
2. Accent only degrees 1, b3, 5, and b7.
3. Accent each landing after a crossing, then remove it.
4. Play the same 36 notes as straight sixteenths.
5. Start at fret 17 or at a randomly chosen rung.

These are Level 23 variations, not separate levels.

Practice log and mastery gate

Date	BPM	Start stroke	Clean route?	Crossing / pulse note	Next target

ADVANCE AFTER: two clean days with both pick starts, true triplets, silent shifts, and correct key naming.